

SQL Exercise - Advanced concepts

Exercise 1: Ranking and Window Functions

```
CREATE DATABASE RetailStore;

USE RetailStore;

DROP TABLE IF EXISTS Products;

CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(100),
    Category VARCHAR(50),
    Price DECIMAL(10,2)
);

INSERT INTO Products VALUES
(1, 'Laptop A', 'Electronics', 80000),
(2, 'Laptop B', 'Electronics', 75000),
(3, 'Laptop C', 'Electronics', 75000),
(4, 'Mobile A', 'Electronics', 50000),
(5, 'Shirt A', 'Clothing', 2000),
(6, 'Shirt B', 'Clothing', 2500),
(7, 'Shirt C', 'Clothing', 2500),
(8, 'Jacket A', 'Clothing', 5000),
(9, 'Book A', 'Books', 500),
(10, 'Book B', 'Books', 700),
(11, 'Book C', 'Books', 700),
(12, 'Book D', 'Books', 1000);

SELECT *
FROM
(
    SELECT
        ProductID,
```

```

    ProductName,

    Category,

    Price,

    ROW_NUMBER() OVER (

        PARTITION BY Category

        ORDER BY Price DESC

    ) AS Row_Num,

    RANK() OVER (

        PARTITION BY Category

        ORDER BY Price DESC

    ) AS Rank_Num,

    DENSE_RANK() OVER (

        PARTITION BY Category

        ORDER BY Price DESC

    ) AS Dense_Rank_Num

FROM Products

) RankedProducts

WHERE Row_Num <= 3

ORDER BY Category, Price DESC;

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 							
	ProductID	ProductName	Category	Price	Row_Num	Rank_Num	Dense_Rank_Num
▶	12	Book D	Books	1000.00	1	1	1
	10	Book B	Books	700.00	2	2	2
	11	Book C	Books	700.00	3	2	2
	8	Jacket A	Clothing	5000.00	1	1	1
	6	Shirt B	Clothing	2500.00	2	2	2
	7	Shirt C	Clothing	2500.00	3	2	2
	1	Laptop A	Electronics	80000.00	1	1	1
	2	Laptop B	Electronics	75000.00	2	2	2
	3	Laptop C	Electronics	75000.00	3	2	2